

Using the previous investigation findings, correlate all events into a single incident. Create: • Attack Timeline • Initial Access • Execution • Persistence (if applicable) • Network Activity • Potential Attacker Objective Separate confirmed evidence from investigative assumptions. Finally assign an overall incident severity.

Tier-1 SOC Incident Correlation Report

Incident Summary

The observed security events correlate into a **single suspected cyber intrusion** involving a privileged account. The sequence indicates a likely compromise of the **Administrator** account, followed by execution of an obfuscated PowerShell command and outbound encrypted communication to an external IP address. This activity aligns with common attacker behavior following successful credential compromise.

Attack Timeline

Time	Event	Correlation
T0	Multiple failed logins (Windows Event ID 4625) from 192.168.10.25 targeting Administrator	Possible brute-force or password guessing attempt
T0 + 5 min	Successful login (Windows Event ID 4624) from the same IP	Potential successful authentication using compromised credentials
Immediately After	Sysmon Event ID 1 - powershell.exe executed with an encoded command	Suspicious command execution following successful login
Shortly After	Sysmon Event ID 3 - powershell.exe established HTTPS connection to 203.0.113.50:443	Possible Command-and-Control (C2), payload retrieval, or data exfiltration

Attack Lifecycle Analysis

1. Initial Access

Confirmed Evidence

- Multiple failed login attempts against the **Administrator** account.
- Successful login from the same IP address (192.168.10.25) five minutes later.
- Same source IP was used for both failed and successful authentication.

Investigative Assumptions

- The attacker may have successfully guessed or obtained the Administrator password.
- The successful login could represent:
 - A brute-force attack succeeding.

- Use of stolen credentials.
- A legitimate user eventually entering the correct password (less likely due to the subsequent malicious activity).

Assessment: Credential compromise is the leading hypothesis, but not conclusively proven from the available evidence alone.

2. Execution

Confirmed Evidence

- Sysmon Event ID 1 shows powershell.exe executed with an encoded command.
- Execution occurred after the successful Administrator login.

Investigative Assumptions

- The encoded command was likely used to:
 - Obfuscate malicious activity.
 - Execute a malware payload.
 - Download additional tools.
 - Evade security monitoring.

Without decoding the command, its exact function remains unknown.

3. Persistence

Confirmed Evidence

- **No persistence mechanisms were observed** in the provided events.

Investigative Assumptions

The attacker may have attempted persistence through techniques such as:

- Registry Run Keys
- Scheduled Tasks
- Windows Services
- WMI Event Subscriptions
- Startup Folder modifications

Additional Sysmon and Windows logs are required to confirm or rule out persistence.

Current Assessment: No evidence of persistence at this stage.

4. Network Activity

Confirmed Evidence

- powershell.exe initiated an outbound HTTPS connection.
- Destination IP: 203.0.113.50
- Destination Port: 443

Investigative Assumptions

The outbound connection may represent:

- Command-and-Control (C2)
- Malware download
- Data exfiltration
- Beaconsing to an attacker-controlled server

Reputation analysis, proxy logs, firewall logs, and packet captures are needed to determine the true purpose of the communication.

Potential Attacker Objective

Confirmed Evidence

The attacker successfully:

- Authenticated to the Administrator account.
- Executed an encoded PowerShell command.
- Established outbound encrypted communication.

Investigative Assumptions

Based on the observed attack chain, the attacker may have intended to:

- Establish remote Command-and-Control.
- Download or execute additional malware.
- Maintain privileged access.
- Conduct reconnaissance.
- Move laterally within the network.
- Exfiltrate sensitive information.
- Prepare for ransomware deployment.

The available evidence supports post-compromise activity but does not conclusively identify the final objective.

Confirmed Evidence vs. Investigative Assumptions

Confirmed Evidence

Multiple failed Administrator logins

Successful Administrator login from same IP

Encoded PowerShell execution

PowerShell initiated outbound HTTPS connection

External IP 203.0.113.50 contacted

No persistence events observed

Investigative Assumptions

Brute-force or password guessing attack

Credentials were compromised

Malicious payload execution

Command-and-Control communication

Malware download or data exfiltration

Persistence may exist but has not yet been identified

Overall Incident Assessment

The correlated events represent a likely attack chain progressing from attempted authentication to successful privileged access, followed by suspicious PowerShell execution and external network communication. While some conclusions (such as credential compromise and C2 activity) remain investigative hypotheses, the combined evidence strongly indicates malicious activity that warrants immediate incident response.

Overall Incident Severity

Critical

Justification

The incident is rated **Critical** because it involves:

- A privileged **Administrator** account.
- Repeated authentication failures followed by a successful login from the same source.
- Execution of an obfuscated (encoded) PowerShell command.
- Outbound encrypted communication initiated by PowerShell.
- Multiple stages of the attack lifecycle occurring in sequence, indicating a high likelihood of an active compromise.

Recommended Immediate Actions

1. Isolate the affected host from the network.
2. Reset or disable the compromised **Administrator** account and rotate privileged credentials.
3. Block communication with 203.0.113.50 pending reputation and forensic analysis.
4. Acquire volatile evidence (memory, running processes, active network connections) if feasible.

5. Collect Windows Security, Sysmon, PowerShell, firewall, proxy, and EDR logs.
6. Hunt across the environment for similar encoded PowerShell executions, failed/successful login patterns, and connections to the same external IP.
7. Escalate the incident to the Tier-2 SOC or Incident Response team for containment, forensic analysis, and eradication.